SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 4
Session#: 4
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will have reviewed unit conversions, algebraic manipulation of exact and inexact numbers, calculating their significant figures after multi-step operations like addition and multiplication. Student will review how to write scienticific notation. Students are given a reference sheet for conversion factors. Students will become adept with atomic particles and subatomic structures like orbitals of electrons.

Opener

Take Attendance

Collaborative Learning Techniques	Clusters
Learning Strategy	Send a Problem
Time	12:00—12:10
Materials	A writing utensil and index cards.
Lecture/Textbook/Old Plans/Slide References	Class roster, Chapter 2: Measurments, Chapter 3: Periodic Table

Definitions

1. **Clusters:** In Clusters, group participants are divided into smaller groups for discussion. They may also be allowed to self-select the small group they want to be in. After discussing the assigned topic, the cluster may report their findings to the large group.

2. **Send a Problem:** Generate a list of problems. Assign a different problem to each student. Give the students a minute to complete the first step of their assigned problem. After the minute, students pass their problem to the students to their right, who then complete the second step of the problem. Continue passing the problem until the steps are completed. After, have the students present their originally-assigned problems with the completed solutions to the entire group.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

Send a Problem (Orbitals Ice Breaker): Generate a list of chemistry problems of varying difficulty. Assign each student a different problem to solve. Give students one minute to complete the first step of their assigned problem. After the minute, each student passes their problem to the group on their right. The receiving group then completes the next step of the problem. This continues until all steps are completed. Finally, each student presents their originally-assigned problem along with the completed solution to the entire class.

1. The SI leader pairs up students based on proximity seating. If a student does not have a partner, the leader joins them.
2. Each pair is assigned a small chemistry task (e.g., determining concentration, calculating molar mass, predicting reaction products) suitable for a first step in a multi-step problem.
3. Pairs have two minutes to decide on their approach and collect any necessary tools or reference materials provided by the leader.
4. Pairs complete the first step of their problem, recording results clearly, and then pass the problem and any physical materials (if applicable) to the next group.
5. The next group evaluates the previous step and completes the subsequent step. If a group identifies an incorrect approach or calculation, they are allowed to correct it for the next step.
6. After all steps are completed, each original problem owner presents the full, completed solution to the class, highlighting any corrections or insights from peer contributions.
7. The SI leader facilitates discussion on problem-solving strategies, peer collaboration, and common chemistry misconceptions.

Students engage collaboratively, seeing multiple approaches to the same problem, learning from their peers, and practicing stepwise problem-solving on a shared "centralized canvas" of the problem.

Content (fully worked out problems & answers) continued...

(1) Calculate the atomic mass of carbon given its common isotopes: ^{12}C has mass 12.000000 amu (98.930%) and ^{13}C has mass 13.003355 amu (1.070%).

a. 12.001 amu

b. 12.01 amu

c. 12.10 amu

d. 13.003 amu

Solution:
Convert abundances to decimals: 0.9893 and 0.0107.

$$\text{atomic mass} = (12.000000)(0.9893) + (13.003355)(0.0107)$$

Compute each term:

$$(12.000000)(0.9893) = 11.87$$

$$(13.003355)(0.0107) = 0.1391$$

$$\text{atomic mass} = 11.87 + 0.1391 = 12.01 \text{ amu}$$

Answer: b. 12.01 amu

Content (fully worked out problems & answers) continued...

(2) Calculate the atomic mass of chlorine given its common isotopes: ^{35}Cl mass = 34.96885 amu (75.78%), ^{37}Cl mass = 36.96590 amu (24.22%).

a. 35.00 amu

b. 35.178 amu

c. 35.45 amu

d. 36.000 amu

Solution:
Convert abundances: 0.7578 and 0.2422.

$$\text{atomic mass} = (34.96885)(0.7578) + (36.96590)(0.2422)$$

Compute each term:

$$(34.96885)(0.7578) = 26.50$$

$$(36.96590)(0.2422) = 8.950$$

$$\text{atomic mass} = 26.50 + 8.950 = 35.45 \text{ amu}$$

Answer: c. 35.45 amu

Content (fully worked out problems & answers) continued...

(3) Calculate the atomic mass of magnesium using three common isotopes: ^{24}Mg mass = 23.985042 amu (78.99%), ^{25}Mg mass = 24.985837 amu (10.00%), ^{26}Mg mass = 25.982593 amu (11.01%).

a. 24.31 amu

b. 24.000 amu

c. 25.000 amu

d. 23.985 amu

Solution:
Convert abundances: 0.7899, 0.1000, 0.1101.

$$\text{atomic mass} = (23.985042)(0.7899) + (24.985837)(0.1000) + (25.982593)(0.1101)$$

Compute each term:

$$(23.985042)(0.7899) = 18.95$$

$$(24.985837)(0.1000) = 2.499$$

$$(25.982593)(0.1101) = 2.861$$

$$\text{atomic mass} = 18.95 + 2.499 + 2.861 = 24.31 \text{ amu}$$

Answer: a. 24.31 amu

Content (fully worked out problems & answers) continued...

(4) Calculate the atomic mass of silver from its two stable isotopes: ^{107}Ag mass = 106.90509 amu (51.839%), ^{109}Ag mass = 108.90476 amu (48.161%).

a. 107.000 amu

b. 107.868 amu

c. 108.000 amu

d. 106.905 amu

Solution:
Convert abundances: 0.51839 and 0.48161.

$$\text{atomic mass} = (106.90509)(0.51839) + (108.90476)(0.48161)$$

Compute each term:

$$(106.90509)(0.51839) = 55.427$$

$$(108.90476)(0.48161) = 52.441$$

$$\text{atomic mass} = 55.427 + 52.441 = 107.868 \text{ amu}$$

Answer: b. 107.868 amu

Content (fully worked out problems & answers) continued...

(5) Calculate the atomic mass of bromine given: ^{79}Br mass = 78.918337 amu (50.690%), ^{81}Br mass = 80.916291 amu (49.310%).

a. 79.000 amu

b. 80.500 amu

c. 79.903 amu

d. 78.918 amu

Solution:
Convert abundances: 0.50690 and 0.49310.

$$\text{atomic mass} = (78.918337)(0.50690) + (80.916291)(0.49310)$$

Compute each term:

$$(78.918337)(0.50690) = 39.999$$

$$(80.916291)(0.49310) = 39.904$$

$$\text{atomic mass} = 39.999 + 39.904 = 79.903 \text{ amu}$$

Answer: c. 79.903 amu

Content (fully worked out problems & answers) continued...

(6) Which group of the periodic table do the alkali metals belong to?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
Alkali metals include Li, Na, K, Rb, Cs, and Fr. These elements are found in the first vertical column of the periodic table, which is Group 1.

Answer: a. Group 1

Content (fully worked out problems & answers) continued...

(7) Calcium (Ca) is classified as which type of element?

a. Alkali metal

b. Alkaline earth metal

c. Halogen

d. Noble gas

Solution:
Calcium (atomic number 20) is found in Group 2 of the periodic table. Group 2 elements are called alkaline earth metals.

Answer: b. Alkaline earth metal

Content (fully worked out problems & answers) continued...

(8) Which period does chlorine (Cl) belong to?

a. Period 2

b. Period 3

c. Period 4

d. Period 5

Solution:
Chlorine (atomic number 17) has its outer electrons in the 3rd energy level, so it belongs to Period 3.

Answer: b. Period 3

(9) Which of the following is a noble gas?

a. Oxygen (O)

b. Fluorine (F)

c. Neon (Ne)

d. Sodium (Na)

Solution:
Noble gases are found in Group 18. Neon (Ne) is in Group 18, while oxygen and fluorine are nonmetals in Groups 16 and 17, and sodium is an alkali metal in Group 1.

Answer: c. Neon (Ne)

Content (fully worked out problems & answers) continued...

(10) Which element is a metalloid?

a. Boron (B)

b. Magnesium (Mg)

c. Argon (Ar)

d. Bromine (Br)

Solution:
Metalloids have properties of both metals and nonmetals. Boron (B) lies along the staircase line and is classified as a metalloid.

Answer: a. Boron (B)

Content (fully worked out problems & answers) continued...

(11) Which group contains the halogens?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
The halogens include fluorine, chlorine, bromine, iodine, and astatine. These are located in Group 17 of the periodic table.

Answer: c. Group 17

Content (fully worked out problems & answers) continued...

(12) Which of the following is a transition metal?

a. Iron (Fe)

b. Sodium (Na)

c. Sulfur (S)

d. Neon (Ne)

Solution:
Transition metals are found in Groups 3–12 of the periodic table. Iron (Fe) is in Group 8, making it a transition metal.

Answer: a. Iron (Fe)

Content (fully worked out problems & answers) continued...

(13) Which element is in Group 2, Period 4?

a. Calcium (Ca)

b. Magnesium (Mg)

c. Beryllium (Be)

d. Strontium (Sr)

Solution:
Group 2 elements are alkaline earth metals. In Period 4, the Group 2 element is Calcium (Ca).

Answer: a. Calcium (Ca)

Content (fully worked out problems & answers) continued...

(14) Which of the following elements is a nonmetal?

a. Aluminum (Al)

b. Phosphorus (P)

c. Iron (Fe)

d. Calcium (Ca)

Solution:
Phosphorus (P) is a nonmetal located in Group 15. The others are metals: aluminum (metal), iron (transition metal), and calcium (alkaline earth metal).

Answer: b. Phosphorus (P)

Content (fully worked out problems & answers) continued...

(15) Which element is in Period 2, Group 18?

a. Helium (He)

b. Neon (Ne)

c. Argon (Ar)

d. Krypton (Kr)

Solution:
Period 2 corresponds to the second row of the periodic table. In Group 18 of this row, the element is Neon (Ne).

Answer: b. Neon (Ne)

Content (fully worked out problems & answers) continued...

(16) Write both the longhand and shorthand (noble-gas) electron configurations for neutral chromium (Cr).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

Solution:
Chromium, Cr, has atomic number $Z = 24$ so there are 24 electrons in the neutral atom. Fill orbitals in order (Aufbau) but note the well-known exception for Cr: a half-filled d subshell+a single s electron is more stable than $4s^2 3d^4$. Thus electrons arrange as:
Longhand (write from lowest to highest energy):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$$

Check electron count: $2 + 2 + 6 + 2 + 6 + 1 + 5 = 24$.
Shorthand: the core through 3p is argon: [Ar]. So shorthand is: [Ar]$4s^1 3d^5$.

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; [Ar]$4s^1 3d^5$

Content (fully worked out problems & answers) continued...

(17) Write both the longhand and shorthand electron configurations for Fe^{3+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

Solution:
Neutral iron, Fe, has $Z = 26$ (26 electrons). Fe^{3+} has lost 3 electrons, so electrons remaining $= 26 - 3 = 23$. When removing electrons from transition metals remove $4s$ electrons before $3d$. Start from neutral Fe configuration: neutral Fe often written [Ar]$4s^2 3d^6$. Remove two $4s$ electrons first (-¿ $4s^0 3d^6$), then remove one $3d$ electron (-¿ $3d^5$). So the result is:
Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ Check count: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: c. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(18) Write both the longhand and shorthand electron configurations for sulfide ion, S^{2-}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^4$; Shorthand: [Ne]$3s^2 3p^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6$; Shorthand: [Ar]

c. Longhand: $1s^2 2s^2 2p^6 3s^1 3p^6$; Shorthand: [Ne]$3s^1 3p^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^5$; Shorthand: [Ne]$3s^2 3p^5$

Solution:
Neutral sulfur S has $Z = 16$ (16 electrons). S^{2-} gains 2 electrons $\Rightarrow$ total electrons $= 16 + 2 = 18$. The configuration for 18 electrons is the same as argon. Longhand filling:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$$

Check count: $2 + 2 + 6 + 2 + 6 = 18$.
Shorthand: the core is argon, so [Ar]. (One can also write [Ne]$3s^2 3p^6$ but [Ar] is simplest.)

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6$; [Ar]

Content (fully worked out problems & answers) continued...

(19) Write both the longhand and shorthand electron configurations for neutral copper (Cu).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; Shorthand: [Ar]$3d^{10} 4s^1$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$; Shorthand: [Ar]$4s^2 3d^9$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{11}$; Shorthand: [Ar]$3d^{11}$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^9$; Shorthand: [Ar]$4s^1 3d^9$

Solution:
Copper, Cu, $Z = 29$ so neutral Cu has 29 electrons. The observed (and most stable) configuration is an exception to the naive Aufbau filling: a filled d subshell ($3d^{10}$) plus a single $4s$ electron is favored. Thus longhand:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$$

Check count: $2 + 2 + 6 + 2 + 6 + 10 + 1 = 29$.
Shorthand: core through 3p is argon, so [Ar]$3d^{10} 4s^1$. Note option (a) orders as $3d^{10} 4s^1$ but writing as [Ar]$4s^1 3d^{10}$ is also seen; the conventional shorthand groups the d after the noble gas core.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; [Ar]$3d^{10} 4s^1$

Content (fully worked out problems & answers) continued...

(20) Write both the longhand and shorthand electron configurations for Mn^{2+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^4$; Shorthand: [Ar]$4s^1 3d^4$

Solution:
Manganese, Mn, has $Z = 25$ (25 electrons) in the neutral atom. Mn^{2+} has lost 2 electrons $\Rightarrow$ electrons remaining $= 25 - 2 = 23$. Neutral Mn is commonly written [Ar]$4s^2 3d^5$. Remove the two $4s$ electrons first when forming cations from transition metals, leaving $4s^0 3d^5$. So the resulting configuration is:
Longhand: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$ Check: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(22) What is an electron orbital?

a. A circular path around the nucleus like planets orbiting the sun

b. A three-dimensional region where there is a high probability of finding an electron

c. A fixed energy value without any spatial meaning

d. A positively charged particle in the nucleus

Solution:
An orbital is not a fixed path (like planets), but rather a mathematical description of the region in space where an electron is most likely to be found. It is based on quantum mechanics and describes electron probability distributions.

Answer: b. A three-dimensional region where there is a high probability of finding an electron

Content (fully worked out problems & answers) continued...

(23) Which statement best describes the difference between an orbital and an orbit?

a. Both mean the same thing: a circular path around the nucleus

b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

c. An orbital is smaller than an orbit

d. An orbit describes the nucleus, while an orbital describes protons

Solution:
Bohr's model described electrons in fixed circular paths called orbits. Modern quantum mechanics replaces this idea with orbitals, which are probability regions where electrons are most likely found. Orbitals are not fixed paths, but probability clouds.

Answer: b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

Content (fully worked out problems & answers) continued...

(24) Draw the electron diagram (orbital boxes) for neutral oxygen (O). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Oxygen has 8 electrons. Filling orbitals using Aufbau/Hund/Pauli:

2p	↑	↑	↑
2s	↑↓		
1s	↑↓		

Answer: $1s^2\ 2s^2\ 2p^4$; 2 unpaired electrons in 2p orbitals.

Content (fully worked out problems & answers) continued...

(25) Draw the electron diagram (orbital boxes) for the chloride ion, Cl^-. Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. All orbitals fully filled:

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: Fully filled orbitals; closed shell.

Content (fully worked out problems & answers) continued...

(26) Draw the electron diagram (orbital boxes) for neutral iron (Fe). Place arrows (↑, ↓) to indicate electrons. Use Aufbau, Hund, and Pauli. Pay attention to how 4s and 3d are filled and later how electrons are paired/unpaired in 3d.

Empty orbital diagram (TikZ)

3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Solution: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$:

3d	↑ ↑ ↑ ↑ ↑↓
4s	↑↓
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Answer: 4 unpaired electrons in 3d.

Content (fully worked out problems & answers) continued...

(27) Draw the electron diagram (orbital boxes) for neutral copper (Cu). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Solution: Cu has electron configuration [Ar] $3d^{10}$ $4s^{1}$:

3d	↑↓ ↑↓ ↑↓ ↑↓ ↑↓
4s	↑
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Answer: $4s^{1}$ $3d^{10}$; 1 unpaired electron in 4s.

Content (fully worked out problems & answers) continued...

(28) Draw the electron diagram (orbital boxes) for the sulfide ion, S^{2-}. Place arrows (↑, ↓) to indicate electrons. Place arrows consistent with Aufbau, Hund, and Pauli.

Empty orbital diagram (TikZ)

3p	☐	☐	☐	
3s	☐			
2p	☐	☐	☐	Fill: 18 electrons; closed shell is expected.
2s	☐			
1s	☐			

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (like Ar):

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: All orbitals filled; closed-shell configuration.

Content (fully worked out problems & answers) continued...

(29) A tectonic plate moves 860 km per day. What is its speed in m/s?

a. 0.995 m/s

b. 9.95 m/s

c. 95.9 m/s

d. 0.0959 m/s

Answer:

$$\text{Speed} = \frac{860\ \cancel{\text{km}}}{\cancel{\text{day}}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \times \frac{1\ \cancel{\text{day}}}{86400\ \text{s}} = 9.95\ \text{m/s}$$

Content (fully worked out problems & answers) continued...

(30) Convert 750 mL to in^3. (Use 1 in = 2.54 cm)

a. 4.58 in^3

b. 45.8 in^3

c. 458 in^3

d. 0.458 in^3

Answer:

$$750 \text{ mL} = 750 \text{ cm}^3, \quad 1 \text{ in}^3 = (2.54 \text{ cm})^3 = 16.387 \text{ cm}^3$$

$$\text{Volume} = \frac{750 \text{ cm}^3}{16.387 \text{ cm}^3/\text{in}^3} = 45.8 \text{ in}^3$$

Content (fully worked out problems & answers) continued...

(31) Convert 88.0 kPa to atm. (Use 1 atm = 101.325 kPa)

a. 0.868 atm

b. 0.880 atm

c. 1.15 atm

d. 0.815 atm

Answer:

$$\text{Pressure} = \frac{88.0\ \cancel{\text{kPa}}}{1} \times \frac{1\ \text{atm}}{101.325\ \cancel{\text{kPa}}} = 0.868\ \text{atm}$$

Content (fully worked out problems & answers) continued...

(32) Convert 4.6 g/L to mg/dL.

a. 0.46 mg/dL

b. 4.6 mg/dL

c. 46 mg/dL

d. 460 mg/dL

Answer:

$$\frac{4.6\ \cancel{g}}{\cancel{L}} \times \frac{1000\ \text{mg}}{1\ \cancel{g}} \times \frac{1\ \text{dL}}{10\ \cancel{L}} = 46\ \text{mg/dL}$$

Content (fully worked out problems & answers) continued...

(33) A block has mass 245.0 g and dimensions 4.50 cm × 3.20 cm × 2.10 cm. Find its density in kg/m^3.

a. 8.10×10^2 kg/m^3

b. 8.10×10^3 kg/m^3

c. 8.10×10^4 kg/m^3

d. 8.10×10^5 kg/m^3

Answer:

$$V = (4.50)(3.20)(2.10)\ \text{cm}^3 = 30.2\ \text{cm}^3$$

$$\rho = \frac{245.0\ \text{g}}{30.2\ \text{cm}^3} = 8.10\ \text{g/mL} = 8.10\ \frac{\text{g}}{\text{cm}^3}$$

$$\text{Density} = \left(8.10\ \frac{\text{g}}{\text{cm}^3}\right) \times \frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{10^6\ \text{cm}^3}{1\ \text{m}^3} = 8.10 \times 10^3\ \text{kg/m}^3$$

Content (fully worked out problems & answers) continued...

(34) A 0.500 g sample of NaCl dissolves in 250.0 mL of water. What is the concentration in g/L?

a. 0.125 g/L

b. 2.00 g/L

c. 1.25 g/L

d. 0.250 g/L

Answer:

$$\text{Concentration} = \frac{0.500 \text{ g}}{0.250 \text{ L}} = 2.00 \text{ g/L}$$

Content (fully worked out problems & answers) continued...

(35) Convert 987.0 mbar to Pa.

a. 9.870 Pa

b. 9.87×10^2 Pa

c. 9.87×10^4 Pa

d. 9.87×10^5 Pa

Answer:

$$987.0 \text{ mbar} \times \frac{100 \text{ Pa}}{1 \text{ mbar}} = 9.870 \times 10^4 \text{ Pa}$$

Content (fully worked out problems & answers) continued...

(36) A sample occupies 250.0 mL and has a mass of 0.890 g. Calculate its density.

a. 3.56×10^{-3} g/mL

b. 0.890 g/mL

c. 3.56 g/mL

d. 35.6 g/mL

Answer:

$$\rho = \frac{0.890 \text{ g}}{250.0 \text{ mL}} = 3.56 \times 10^{-3} \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(37) A beaker contains 3.50 L of solution. Express this volume in cubic decimeters (dm^3) using unit factors.

a. 0.00350 dm^3

b. 3.50 dm^3

c. 35.0 dm^3

d. 350 dm^3

Answer:

$$3.50 \text{ L} \times \frac{10^{-3} \text{ dm}^3}{1 \text{ L}} = 0.00350 \text{ dm}^3$$

Content (fully worked out problems & answers) continued...

(38) A student measures 0.00250 kg of a substance. Express this in grams.

a. 0.00250 g

b. 0.0250 g

c. 2.50 g

d. 250 g

Answer:

$$0.00250 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 2.50 \text{ g}$$

Content (fully worked out problems & answers) continued...

(39) A 5.00 mL sample of water weighs 4.98 g. What is its density?

a. 0.996 g/mL

b. 1.00 g/mL

c. 1.02 g/mL

d. 1.10 g/mL

Answer:

$$\rho = \frac{4.98 \text{ g}}{5.00 \text{ mL}} = 0.996 \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(40) Which of the following is an isotope of carbon?

a. ^{12}C

b. ^{14}C

c. Both a and b

d. ^{16}O

Solution:
Isotopes are atoms of the same element with different mass numbers (different neutrons). Both ^{12}C and ^{14}C are carbon isotopes.

Answer: c. Both a and b

Content (fully worked out problems & answers) continued...

(41) Which subatomic particle determines the identity of an element?

a. Proton

b. Electron

c. Neutron

d. Nucleus

Solution:
The number of protons (atomic number) defines the element. Changing the number of protons changes the element.

Answer: a. Proton

Content (fully worked out problems & answers) continued...

(42) An atom of sodium (Na) loses one electron. What is the resulting ion?

a. Na^{-}

b. Na^{+}

c. Na^{2+}

d. Neutral Na

Solution:
Losing one electron creates a positive charge. Sodium forms Na^{+} when it loses one electron.

Answer: b. Na^{+}

Content (fully worked out problems & answers) continued...

(43) Which of the following are isotopes of the same element?

a. ^{1}H and ^{2}H

b. ^{12}C and ^{14}N

c. ^{16}O and ^{18}F

d. ^{23}Na and ^{24}Mg

Solution:
Isotopes are atoms with the same number of protons but different neutrons. ^{1}H and ^{2}H are hydrogen isotopes.

Answer: a. ^{1}H and ^{2}H

Content (fully worked out problems & answers) continued...

(46) Which type of bond is expected between Na and Cl in NaCl?

a. Nonpolar covalent

b. Polar covalent

c. Ionic

d. Metallic

Solution:
Electronegativity difference between Na (0.9) and Cl (3.0) is about 2.1, which is large. A difference greater than ~ 2.0 typically indicates ionic bonding. Thus, Na transfers an electron to Cl, forming an ionic bond.

Answer: c. Ionic

Content (fully worked out problems & answers) continued...

(47) Which of the following has the highest electronegativity?

a. Lithium

b. Carbon

c. Fluorine

d. Oxygen

Solution:
Electronegativity increases across a period and decreases down a group. Fluorine is the most electronegative element (value ≈ 4.0), higher than oxygen (3.5), carbon (2.5), and lithium (1.0).

Answer: c. Fluorine

Content (fully worked out problems answers) continued...

(48) What is the correct name for MgF_2?

a. Magnesium difluoride

b. Magnesium fluoride

c. Magnesium(II) fluoride

d. Monomagnesium fluoride

Solution:
Magnesium is Group 2 (+2). Fluoride is -1, two are needed to balance. The correct name is magnesium fluoride.

Answer: b. Magnesium fluoride

Content (fully worked out problems & answers) continued...

(49) Which is the correct name for Cu_2O?

a. Copper(I) oxide

b. Copper(II) oxide

c. Dicopper monoxide

d. Copper oxide

Solution:
Oxygen is -2. Two Cu atoms balance it, so each Cu is +1. Correct name: Copper(I) oxide.

Answer: a. Copper(I) oxide

Content (fully worked out problems & answers) continued...

(50) What is the formula for lithium sulfide?

a. LiS

b. Li_2S

c. LiS_2

d. Li_2S_3

Solution:
Lithium is +1, sulfide is -2. Two Li^+ are required for one S^{2-}. Formula: Li_2S.

Answer: b. Li_2S

Content (fully worked out problems & answers) continued...

(51) What is the correct name for PbO_2?

a. Lead(II) oxide

b. Lead(IV) oxide

c. Lead oxide

d. Dilead dioxide

Solution:
Oxygen is -2, two atoms total -4. Pb must be +4. Correct name: Lead(IV) oxide.

Answer: b. Lead(IV) oxide

Content (fully worked out problems & answers) continued...

(52) Which of the following is the correct formula for strontium chloride?

a. SrCl

b. Sr_2Cl

c. $SrCl_2$

d. Sr_2Cl_3

Solution:
Strontium is Group 2 (+2). Chloride is -1, so two Cl^- are needed. Formula: $SrCl_2$.

Answer: c. $SrCl_2$

Content (fully worked out problems answers) continued...

(53) What is the correct name for $FeCl_3$?

a. Iron chloride

b. Iron(II) chloride

c. Iron(III) chloride

d. Ferrous chloride

Solution:
Each Cl is -1, three Cl give -3 total, so Fe must be +3. Use Roman numeral: Iron(III) chloride.

Answer: c. Iron(III) chloride

Content (fully worked out problems & answers) continued...

(54) Which is the correct name for CoO?

a. Cobalt oxide

b. Cobalt(II) oxide

c. Cobalt(III) oxide

d. Dicobalt monoxide

Solution:
O is -2, one Co balances with +2. That gives cobalt in the +2 oxidation state: Cobalt(II) oxide.

Answer: b. Cobalt(II) oxide

Content (fully worked out problems & answers) continued...

(55) What is the correct name for Cu_2S?

a. Copper sulfide

b. Copper(II) sulfide

c. Copper(I) sulfide

d. Dicopper monosulfide

Solution:
Sulfide is -2. Two Cu atoms give +2 total so each Cu is +1. Name uses Roman numeral: Copper(I) sulfide.

Answer: c. Copper(I) sulfide

Content (fully worked out problems answers) continued...

(56) What is the correct name for $Zn(NO_3)_2$?

a. Zinc nitrate

b. Zinc dinitrate

c. Zinc(II) nitrite

d. Zinc nitrite

Solution: Nitrate is -1; two nitrates give -2, so Zn must be $+2$. Name: zinc (common oxidation state $+2$) with nitrate.

Answer: a. Zinc nitrate

Content (fully worked out problems answers) continued...

(57) Which is the correct formula for silver sulfate?

a. $AgSO_4$

b. Ag_2SO_4

c. $Ag(SO_4)_2$

d. $Ag_2(SO_3)$

Solution: Sulfate is -2 and Ag is $+1$, so two Ag^+ are needed per sulfate: formula Ag_2SO_4.

Answer: b. Ag_2SO_4

Content (fully worked out problems answers) continued...

(58) What is the correct name for $Cd(OH)_2$?

a. Cadmium hydroxide

b. Cadmium(II) hydroxide

c. Cadmium dihydroxide

d. Cadmium(I) hydroxide

Solution: Hydroxide is -1; two OH^- give -2, so Cd is $+2$. Proper systematic name: cadmium(II) hydroxide (commonly "cadmium hydroxide").

Answer: b. Cadmium(II) hydroxide

Content (fully worked out problems answers) continued...

(59) Which is the correct name for $Ga_2(CO_3)_3$?

a. Gallium carbonate

b. Gallium(II) carbonate

c. Gallium(III) carbonate

d. Digallium tricarbonate

Solution: Carbonate is -2; three give -6, two Ga supply $+6$ total so each Ga is $+3$. Name: gallium(III) carbonate (commonly "gallium carbonate").

Answer: c. Gallium(III) carbonate

Content (fully worked out problems answers) continued...

(60) What is the formula for ammonium phosphate?

a. $(NH_4)PO_4$

b. $(NH_4)_2PO_4$

c. $(NH_4)_3PO_4$

d. NH_4PO_4

Solution: Phosphate is -3; ammonium is $+1$ so three NH_4^+ are needed to balance one PO_4^{3-}: $(NH_4)_3PO_4$.

Answer: c. $(NH_4)_3PO_4$

Content (fully worked out problems answers) continued...

(61) What is the correct name for $Al_2(SO_4)_3$?

a. Aluminum sulfate

b. Aluminum(III) sulfate

c. Dialuminum trisulfate

d. Aluminum sulfite

Solution: Sulfate is -2; three sulfates give -6, two Al give $+6$ so Al is $+3$. Common name: aluminum sulfate (aluminum(III) sulfate is equivalent).

Answer: a. Aluminum sulfate

Content (fully worked out problems answers) continued...

(62) Which is the correct formula for zinc acetate?

a. $Zn(C_2H_3O_2)$

b. $Zn(C_2H_3O_2)_2$

c. $Zn_2(C_2H_3O_2)_3$

d. $Zn_2(C_2H_3O_2)_2$

Solution: Acetate is -1; Zn is $+2$, so two acetates are needed: $Zn(C_2H_3O_2)_2$.

Answer: b. $Zn(C_2H_3O_2)_2$

Content (fully worked out problems answers) continued...

(63) What is the correct name for $Ag_3(PO_4)$?

a. Silver phosphate

b. Silver(III) phosphate

c. Trisilver phosphate

d. Silver(I) phosphate

Solution: Phosphate is -3, Ag is $+1$ so three Ag^+ balance one PO_4^{3-}. Name: silver phosphate (silver(I) phosphate is equivalent).

Answer: a. Silver phosphate

Content (fully worked out problems answers) continued...

(64) Which is the correct formula for cadmium sulfite?

a. $CdSO_3$

b. $Cd_2(SO_3)_3$

c. $Cd(SO_3)_2$

d. $Cd_3(SO_3)_2$

Solution: Sulfite is -2; Cd is typically $+2$, so a 1:1 formula $CdSO_3$.

Answer: a. $CdSO_3$

Content (fully worked out problems answers) continued...

(65) What is the correct name for $Ga(NO_2)_3$?

a. Gallium nitrate

b. Gallium nitrite

c. Gallium(III) nitrite

d. Gallium trinitrate

Solution: Nitrite is -1; three nitrites give -3, Ga must be $+3$. Proper name: gallium(III) nitrite (commonly "gallium nitrite").

Answer: c. Gallium(III) nitrite

Content (fully worked out problems answers) continued...

(66) What is the formula for ammonium chromate?

a. $(NH_4)CrO_4$

b. $(NH_4)_2CrO_4$

c. $(NH_4)_3CrO_4$

d. NH_4CrO_4

Solution: Chromate is CrO_4^{2-}; ammonium is $+1$, so two ammoniums are required: $(NH_4)_2CrO_4$.

Answer: b. $(NH_4)_2CrO_4$

Content (fully worked out problems answers) continued...

(67) Which is the correct name for $Zn(CN)_2$?

a. Zinc cyanide

b. Zinc dicyanide

c. Zinc(II) cyanide

d. Zinc cyanate

Solution: Cyanide is -1; two CN^- balance Zn^{2+} giving $Zn(CN)_2$. Name: zinc cyanide (or zinc(II) cyanide).

Answer: a. Zinc cyanide

Content (fully worked out problems answers) continued...

(68) What is the correct name for $Al(OH)_3$?

a. Aluminum hydroxide

b. Aluminum(III) hydroxide

c. Trialuminum hydroxide

d. Aluminum dihydroxide

Solution: Hydroxide is -1; three OH^- give -3, so Al is $+3$. Name: aluminum hydroxide (aluminum(III) hydroxide is explicit).

Answer: a. Aluminum hydroxide

Content (fully worked out problems answers) continued...

(69) Which is the correct formula for silver acetate?

a. $AgC_2H_3O_2$

b. $Ag(C_2H_3O_2)_2$

c. $Ag_2(C_2H_3O_2)$

d. $Ag_2(C_2H_3O_2)_2$

Solution: Acetate is singly charged (-1) and Ag is $+1$, so one acetate per Ag: $AgC_2H_3O_2$.

Answer: a. $AgC_2H_3O_2$

Content (fully worked out problems answers) continued...

(70) What is the correct name for $Cd_3(PO_4)_2$?

a. Cadmium phosphate

b. Cadmium(III) phosphate

c. Tricadmium diphosphate

d. Cadmium(II) phosphate

Solution: Phosphate is -3, two phosphates give -6, three Cd give $+6$, so each Cd is $+2$. Name: cadmium(II) phosphate (commonly "cadmium phosphate").

Answer: d. Cadmium(II) phosphate

Content (fully worked out problems answers) continued...

(71) Which is the correct formula for gallium hydroxide?

a. GaOH

b. $Ga(OH)_2$

c. $Ga(OH)_3$

d. $Ga_2(OH)_3$

Solution: Ga is +3 and hydroxide is −1, so three OH^- required: $Ga(OH)_3$.

Answer: c. $Ga(OH)_3$

Content (fully worked out problems answers) continued...

(72) What is the correct name for $ZnSO_3$?

a. Zinc sulfate

b. Zinc sulfite

c. Zinc(II) sulfite

d. Zinc sulfur trioxide

Solution: Sulfite is SO_3^{2-} and Zn is +2, so formula $ZnSO_3$ is 1:1. Name: zinc sulfite (systematically zinc(II) sulfite).

Answer: c. Zinc(II) sulfite

Content (fully worked out problems answers) continued...

(73) Which is the correct formula for ammonium nitrate?

a. $(NH_4)NO_2$

b. $(NH_4)NO_3$

c. $(NH_4)_2NO_3$

d. NH_4NO_2

Solution: Nitrate is NO_3^- and ammonium is NH_4^+; they combine 1:1 giving $(NH_4)NO_3$ (often written NH_4NO_3).

Answer: b. $(NH_4)NO_3$

Content (fully worked out problems answers)continued...

(74) What is the correct name for Ag_2CO_3?

a. Silver carbonate

b. Silver(II) carbonate

c. Disilver carbonate

d. Silver(I) carbonate

Solution: Carbonate is -2; Ag is $+1$, so two Ag^+ balance CO_3^{2-}: Ag_2CO_3. Name: silver carbonate (silver(I) carbonate is equivalent).

Answer: a. Silver carbonate

Content (fully worked out problems answers) continued...

(75) Which is the correct formula for aluminum nitrite?

a. $Al(NO_2)$

b. $Al(NO_2)_2$

c. $Al(NO_2)_3$

d. $Al_2(NO_2)_3$

Solution: Nitrite is NO_2^-; Al is +3, so three nitrite ions are needed: $Al(NO_2)_3$.

Answer: c. $Al(NO_2)_3$

Content (fully worked out problems & answers) continued...

(76) Start from the Lewis dot of a neutral sodium atom (Na) and draw the Lewis dot structure for the sodium ion, Na^+. Indicate what happens to the valence electron and the resulting charge.

Solution:
Neutral Na has 1 valence electron (one dot). Removing that electron gives Na^+ with no valence dots and a +1 charge.

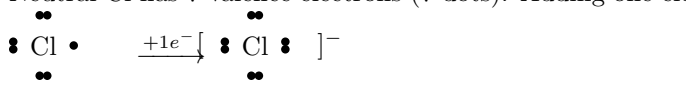
$$\text{Na} \bullet \quad \xrightarrow{-1e^-} [\quad \text{Na} \quad]^+$$

Answer: Na^+ — electron removed, no valence dots, charge +1.

Content (fully worked out problems & answers) continued...

(77) Given the Lewis dot for a neutral chlorine atom, draw the Lewis dot structure for the chloride ion, Cl^-. Show where the added electron is and indicate the charge.

Solution:
Neutral Cl has 7 valence electrons (7 dots). Adding one electron gives 8 (octet) and a -1 charge.

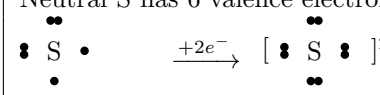


Answer: Cl^- — octet completed, charge -1.

Content (fully worked out problems & answers) continued...

(78) Start from the Lewis dot of a neutral sulfur atom (S) and draw the Lewis dot structure for the sulfide ion, S^{2-}. Show electron counting and charge.

Solution:
Neutral S has 6 valence electrons. Adding 2 electrons yields 8 (octet) and a -2 charge.



Answer: S^{2-} — octet, charge -2.

Content (fully worked out problems & answers) continued...

(79) Given the Lewis dot of a neutral magnesium atom (Mg), draw the Lewis structure for Mg^{2+}. Indicate removal of valence electrons and the resulting charge.

Solution:
Neutral Mg has 2 valence electrons (two dots). Removing both gives Mg^{2+} with no valence dots and a +2 charge.

$$\text{Mg}\,\bullet\bullet \xrightarrow{-2e^-} [\ \text{Mg}\ ]^{2+}$$

Answer: Mg^{2+} — two electrons removed, charge +2.

Content (fully worked out problems & answers) continued...

(80) Start from the Lewis dot for a neutral copper atom (showing its single 4s valence electron as one dot) and draw the Lewis structure for the cuprous ion, Cu^+. Explain briefly.

Solution:
A common simple Lewis-dot convention places one valence dot for Cu ($4s^1$). Removing that electron gives Cu^+ with no valence dot and charge +1. (Note: d-electrons are not shown in simple Lewis-dot for many transition metals.)

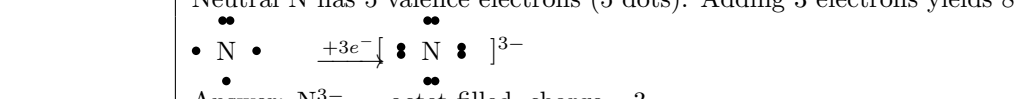
$$\text{Cu} \bullet \quad \xrightarrow{-1e^-} [\quad \text{Cu} \quad]^+$$

Answer: Cu^+ — 4s electron removed, no valence dot, charge +1.

Content (fully worked out problems & answers) continued...

(81) Given the Lewis dot for neutral nitrogen (N), draw the Lewis dot for nitride ion, N^{3-}, and indicate electron counting and charge.

Solution:
Neutral N has 5 valence electrons (5 dots). Adding 3 electrons yields 8 (octet) and a -3 charge.



Answer: N^{3-} — octet filled, charge -3.

Activity 1

Collaborative Learning Techniques	Group Discussion
Learning Strategy	Vocabulary Development
Time	12:10—12:30
Materials	Notes
Lecture/Textbook/Old Plans/Slide References	Chapter 3: Periodic Table

Definition

1. **Group Discussion:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.

2. **Vocabulary Development:** Chunking related terms into meaningful groups can be more helpful than drilling students on exact definitions. Compose a list of key terms from the lecture rangin in levels of specificity, Scramble the terms and then encourage pairs of students to organize the terms into serveral categories that are meaninful to them. Then have them define or give an example of terms where appropriate. Finally, have each pair discuss their categories with the entire group. Get the students to check spelling.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. **Form a Circle:** All students sit or stand in a circle so that everyone can see and interact with each other.
2. **Prepare Terms by Specificity:** Compose a comprehensive list of key terms related to ions, atomic orbitals, molecules, bonds, elements, isotopes, measurements, electron configuration, and VSEPR. Organize terms into levels of specificity:
   - *Level 1 – Broad Concepts:* e.g., "element," "molecule"
   - *Level 2 – Intermediate Concepts:* e.g., "cation," "polar covalent bond," "2p orbital"
   - *Level 3 – Specific Examples:* e.g., "Na+," "H–Cl bond," "sp^2 hybridization"
3. **Scramble and Distribute:** Shuffle the terms and give each student or pair a subset to organize within the specificity levels.
4. **Round-Robin Discussion:** Moving around the circle, each student or pair presents their terms starting from the broadest level to the most specific. For each term, they provide a definition or example.
5. **Peer Verification:** Other students in the circle can suggest corrections, alternative examples, or clarifications. Emphasize discussion rather than right/wrong answers.
6. **Spelling and Accuracy Check:** Ensure terms are spelled correctly and scientifically accurate. The instructor or SI leader monitors and provides guidance.
7. **Reflection:** After all terms are discussed, students reflect on the hierarchical relationships between concepts, noting how broad categories encompass intermediate and specific examples.

Checking for Understanding (general conceptual questions)

1. How do you identify cations and anions, and what processes form them?
2. How can electron configurations predict the chemical behavior and reactivity of elements?
3. What is the difference between an atom, an ion, and an isotope, and how do these differences affect chemical properties?
4. How do orbitals differ from orbits, and why is this distinction important for understanding electron arrangements?
5. How can you determine the shape of a molecule using VSEPR theory?
6. What is the difference between ionic, polar covalent, and nonpolar covalent bonds, and how can you predict which will form?
7. How do electronegativity differences relate to bond polarity and molecular polarity?
8. How do you determine if a molecule is polar or nonpolar overall?
9. How do the number of protons, neutrons, and electrons determine the identity and charge of an atom or ion?
10. How can you organize elements using the periodic table, and what patterns help predict properties?
11. How does atomic and molecular structure influence physical properties such as density or polarity?
12. How do measurements in chemistry, such as mass, volume, and concentration, relate to chemical calculations and formulas?
13. How do isotopes of the same element differ in mass but retain the same chemical behavior?
14. How do specific orbitals (s, p, d, f) relate to electron distribution in atoms and the periodic trends?
15. How do scientific notation and powers of ten help when representing very large or very small quantities in chemistry?

Activity 2

Collaborative Learning Techniques	Jigsaw
Learning Strategy	Boardwork Model (modified)
Time	12:30—12:50
Materials	Colored whiteboard markers, a whiteboard, partitioned questions (to display multiple to multiple groups at a time).
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.

2. **Boardwork Model:** This is a method of organizing boardwork in order to fascilitate an undertsanding of problem-solving strategies. The baord should be divdied into 4 sections: 1.) Prerequisite Knowledge, 2.) Mathematical Steps, 3.) Narrative of the Steps, 4.) Additional Sample Problem. Encourage one student to fill out section 1 on the board. Then encourage 2 students to simultaneously complete section 2 and 3 on the board. Lastly, have another student complete section 4. Encourage the students to use this model when studying outside of the SI session.

Instructions (please provide aDETAILED activity breakdown below (and/or attached)

Objective: Guide students to work independently on a chemistry worksheet, solving problems. After asking a survey for answers completing each piece, students share their solutions and recieve feedback on each step to ensure understanding before proceeding.

Setup:

- Prepare a worksheet divided into sections or "pieces," each containing a set of chemistry problems (e.g., ions, electron configuration, molecular geometry, bonding, measurements).
- Each piece should build upon the previous one to create a logical progression from broad to specific concepts.
- Provide reference materials (periodic table, formulas, or charts) as needed.
- Ensure students have paper, pencils, and optionally colored pens for diagrams or drawings.

Running the Activity:

1. Explain that students will complete the worksheet **piece by piece**, solving each section on their own first.
2. After finishing a piece, students **share their answers** with a partner or small group to check understanding.
3. Students discuss each solution, correcting errors or clarifying concepts before moving to the next piece.
4. Once all students confirm understanding of the current piece, they move to the next section of the worksheet.
5. Continue this process until the entire worksheet is completed.
6. Encourage students to annotate their worksheets with notes from peer discussions to reinforce learning.

Facilitator Role:

- Circulate to monitor student progress and understanding.
- Provide hints or clarification without giving away solutions.
- Ensure students are actively discussing and verifying each piece before moving on.
- Take note of common misunderstandings for future review or debriefing.
- At the end, lead a brief class discussion summarizing key concepts and connections between the worksheet sections.

Checking for Understanding (questions & answers)

Questions:

1. How does solving the worksheet in sequential "pieces" help students build understanding step by step?
2. Why is sharing answers with peers after each piece beneficial for learning chemistry concepts?
3. In what ways does this approach encourage students to actively reflect on their own understanding?
4. How can verifying each piece with a partner prevent misconceptions from carrying forward?
5. How does this method promote independent problem-solving skills while still incorporating collaboration?
6. What strategies can students use to interpret and categorize chemistry terms as they work through the worksheet?
7. How does this sequential, piece-by-piece approach help students make connections between broad and specific concepts (e.g., elements → ions → orbitals → molecules)?
8. How can discussing each piece improve students' ability to explain chemical reasoning clearly?
9. What advantages do students gain from seeing multiple approaches to the same problem or term?
10. How does this model help students practice applying theoretical knowledge to concrete problems?
11. In what ways can the instructor or SI leader use this approach to identify areas where students struggle?
12. How does completing the worksheet in pieces enhance retention of chemistry concepts for students with varying levels of prior knowledge?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:50—12:55
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.

2. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are put into small groups and are asked to write down their confidence on a single notecard without names.
2. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers) continued...

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams